

Harouna Thiam

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EDUCATION

Duke University

Durham, NC

B.S. in Computer Science + Artificial Intelligence & Machine Learning

Expected May 2028

Relevant Coursework: DSA, Computer Systems, Discrete Mathematics, Probability, Distributed Systems, Linear Algebra

Awards: HackDuke Winner, U.S. Congressional Recognition in Machine Learning, Amazon Future Engineer Scholarship,

LinkedIn Possibilities in Tech Scholarship, Jane Street UNBOXED Scholarship, Bradenton Golden Herald Award in Technology

SKILLS

Languages: Python, Java, React, SQL, Kotlin, TypeScript/JavaScript, R, Swift, Go

Infrastructure: AWS (Lambda, S3, Bedrock, SNS), Docker, git, CI/CD, Github Actions, REST, Distributed Systems, Firebase

AI/ML/Data: Lanchain, Vectorstores, Pandas, Prompt Engineering, NumPy, Tensorflow, Codex, Claude Code, Gemini, Cursor

EXPERIENCE

Amazon • Software Development Intern

May. 2025 – Aug. 2025

- Built a serverless cloud workflow using Java and AWS Step Functions to automate JSON data transformation with LLMs
- Reduced JSON processing time by **87%** by using a distributed parallel map state with AWS Bedrock and State Machines
- Designed a callback token mechanism with TypeScript SNS listeners and Lambda gateways to synchronize validation responses
- Implemented an Utterance Validation Framework with handoffs to S3 between workflow states to validate Alexa text-to-speech
- Ensured code quality with automated testing suites (unit, integration, load) with JUnit, integrated into a CI/CD pipeline

Duke Fuqua School of Business • Technical Coordinator & Grader

Apr. 2025 – Current

Courses: Software Tools for Analytics (DECISION 616), Fraud Analytics (DECISION 523Q)

- Built and deployed a course submission autograder using Python, Pandas, Openpyxl. Decreased grading time for TAs by **42%**
- Integrated generative AI prompts across **150+** pages of textbooks to create AI-assisted instructions for graduate coursework
- Facilitated the integration of a new Learning Management System (LMS) for AI-Enhanced graduate student learning
- Overseeing TA team for a course of **400+** students, delegated assignment grading for TAs

ZSuite Technologies • AI/ML Software Engineer Intern

Jun. 2023 – Aug. 2023

- Developed LLM chatbot tool with Python and Lanchain, resulting in increased efficiency of company representative response
- Curated relevant context to an LLM from CSV files through a vector store with Weaviate and RAG, improving query accuracy
- Containerized code and workspace with Docker and Jupyter Notebooks to create a scalable and modular codebase

PROJECTS

HackDuke 'Code for Good' Hackathon • 1st Place App in Sustainability

Feb. 2025

- Developed a full mobile application in a team setting at HackDuke, utilizing flutter to create an interactive daily challenge platform
- Utilized Gemini API to dynamically generate engaging personal challenges and gamified progress for users with LLMs
- Used game design and UI/UX principles to secure **150+** downloads with engaging storytelling and progress tracking

United States Congressional App Challenge • 1st Place App - Florida

Nov. 2023

- Integrated image classification models using TensorFlow to detect invasive species and diagnose plant conditions (mobile)
- Utilized Kotlin coroutines and XML to generate dynamic plant descriptions with OpenAI and asynchronous data flows
- Optimized mobile performance by using TensorFlow Lite for on-device inferences, reducing latency and ensuring offline access

LinkedIn Voyage: LinkedIn Learning

Jun. 2025

- Designed and presented a revamp of LinkedIn Learning's 'Career Charter' feature to LinkedIn engineers and executives at HQ.
- Utilized Google Gemini API and RAG to generate dynamic and chronological course lists with context-aware LLM generation
- Prototyped a recommendation workflow using semantic retrieval and prompt engineering to personalize course paths

PROFESSIONAL DEVELOPMENT

Amazon Future Engineer • One of **400** students internationally selected for early career engineering skills development

LinkedIn Possibilities in Tech • One of **25** students internationally selected for annual skills development with LinkedIn execs/CEO

Walmart Technology Summit • Selected among **100** tech students to explore Walmart HQ and connect with business leaders

Goldman Sachs Engineering Summit • **13%** selected for an intensive multi-month development program with world-class analysts

D.E. Shaw Connect • Invited among high achieving sophomores to NYC to connect with D.E. Shaw engineers and business execs

Management Leadership for Tomorrow (MLT) • Selected for 18-month intensive technology mentorship with various companies

ACTIVITIES & INTERESTS

Alliance for Innovative and Interdisciplinary Computing Education (AiiCE) • National Student Advisory Board Member

ColorStack@Duke & National Society of Black Engineers • Member